

Q1 - 24 January - Shift 1

The weight of a body at the surface of earth is 18 N. The weight of the body at an altitude of 3200 km above the earth's surface is (given, radius of earth $R_e = 6400$ km)

- (1) 9.8 N (2) 4.9 N
(3) 19.6 N (4) 8 N

Space for your notes:

Q2 - 24 January - Shift 2

Given below are two statements:

Statement I: Acceleration due to earth's gravity decreases as you go 'up' or 'down' from earth's surface.

Statement II: Acceleration due to earth's gravity is same at a height 'h' and depth 'd' from earth's surface, if $h = d$.

In the light of above statements, choose the **most appropriate** answer from the options given below

- (1) Statement I is incorrect but statement II is correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Both Statement I and II are correct

Space for your notes:

Q3 - 24 January - Shift 2

If the distance of the earth from Sun is 1.5×10^6 km. Then the distance of an imaginary planet from Sun, if its period of revolution is 2.83 years is:

- (1) 6×10^7 km
- (2) 6×10^6 km
- (3) 3×10^6 km
- (4) 3×10^7 km

Space for your notes:

Q4 - 24 January - Shift 2

Given below are two statements: one is labelled as

Assertion A and the other is labelled as **Reason R**.

Assertion A: A pendulum clock when taken to Mount Everest becomes fast.

Reason R: The value of g (acceleration due to gravity) is less at Mount Everest than its value on the surface of earth.

In the light of the above statements, choose the most appropriate answer from the options given below

- (1) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A**
- (2) Both **A** and **R** are correct and **R** is the correct explanation of **A**
- (3) **A** is not correct but **R** is correct
- (4) **A** is correct but **R** is not correct

Space for your notes:

Q5 - 25 January - Shift 1

Assume that the earth is a solid sphere of uniform density and a tunnel is dug along its diameter throughout the earth. It is found that when a particle is released in this tunnel, it executes a simple harmonic motion. The mass of the particle is 100 g. The time period of the motion of the particle will be (approximately) (take $g = 10 \text{ ms}^{-2}$, radius of earth = 6400 km)

- (1) 24 hours
- (2) 1 hour 24 minutes
- (3) 1 hour 40 minutes
- (4) 12 hours

Space for your notes:

Q6 - 25 January - Shift 1

T is the time period of simple pendulum on the earth's surface. Its time period becomes $x \cdot T$ when taken to a height R (equal to earth's radius) above the earth's surface. Then, the value of x will be:

- (1) 4
- (2) 2
- (3) $\frac{1}{2}$
- (4) $\frac{1}{4}$

Space for your notes:

Q7 - 25 January - Shift 2

A body of mass is taken from earth surface to the height h equal to twice the radius of earth (R_e), the increase in potential energy will be : (g = acceleration due to gravity on the surface of Earth)

- (1) $3mgR_e$
- (2) $\frac{1}{3}mgR_e$
- (3) $\frac{2}{3}mgR_e$
- (4) $\frac{1}{2}mgR_e$

Space for your notes:

Q8 - 25 January - Shift 2

Every planet revolves around the sun in an elliptical orbit :

A. The force acting on a planet is inversely proportional to square of distance from sun.

B. Force acting on planet is inversely proportional to product of the masses of the planet and the sun

C. The centripetal force acting on the planet is directed away from the sun.

D. The square of time period of revolution of planet around sun is directly proportional to cube of semi-major axis of elliptical orbit.

Choose the correct answer from the options given below :

Options :

- (1) A and D only (2) C and D only
(3) B and C only (4) A and C only

Space for your notes:

Q9 - 29 January - Shift 1

Two particles of equal mass 'm' move in a circle of radius 'r' under the action of their mutual gravitational attraction. The speed of each particle will be :

- (1) $\sqrt{\frac{GM}{2r}}$ (2) $\sqrt{\frac{4GM}{r}}$
(3) $\sqrt{\frac{GM}{r}}$ (4) $\sqrt{\frac{GM}{4r}}$

Space for your notes:

Q10 - 29 January - Shift 2

The time period of a satellite of earth is 24 hours. If the separation between the earth and the satellite is decreased to one fourth of the previous value, then its new time period will become.

- (1) 4 hours (2) 6 hours
(3) 12 hours (4) 3 hours

Space for your notes:

Q11 - 30 January - Shift 1

If the gravitational field in the space is given as

$\left(-\frac{K}{r^2}\right)$. Taking the reference point to be at

$r = 2$ cm with gravitational potential $V = 10$ J/kg.

Find the gravitational potential at $r = 3$ cm in SI unit (Given, that $K = 6$ J cm/kg)

- (1) 9 (2) 11
(3) 12 (4) 10

Space for your notes:

Q12 - 31 January - Shift 1

At a certain depth "d" below surface of earth, value of acceleration due to gravity becomes four times that of its value at a height $3R$ above earth surface. Where R is Radius of earth (Take $R = 6400$ km).

The depth d is equal to

- (1) 5260 km
(2) 640 km
(3) 2560 km
(4) 4800 km

Space for your notes:

Q13 - 31 January - Shift 1

Spherical insulating ball and a spherical metallic ball of same size and mass are dropped from the same height. Choose the correct statement out of the following {Assume negligible air friction}

- (1) Time taken by them to reach the earth's surface will be independent of the properties of their materials
- (2) Insulating ball will reach the earth's surface earlier than the metal ball
- (3) Both will reach the earth's surface simultaneously
- (4) Metal ball will reach the earth's surface earlier than the insulating ball.

Space for your notes:

Q14 - 31 January - Shift 2

A body weight W , is projected vertically upwards from earth's surface to reach a height above the earth which is equal to nine times the radius of earth. The weight of the body at that height will be:

- (1) $\frac{W}{91}$
- (2) $\frac{W}{100}$
- (3) $\frac{W}{9}$
- (4) $\frac{W}{3}$

Space for your notes:

Q15 - 01 February - Shift 1

Given below are two statements :

Statement-I: Acceleration due to gravity is different at different places on the surface of earth.

Statement-II: Acceleration due to gravity increases as we go down below the earth's surface.

In the light of the above statements, choose the **correct** answer from the options given below

- (1) Both Statement I and Statement II are true
- (2) Both Statement I and Statement II are false
- (3) Statement I is true but Statement II is false
- (4) Statement I is false but Statement II is true

Space for your notes:

Q16 - 01 February - Shift 1

If earth has a mass nine times and radius twice to the of a planet P. Then $\frac{v_e}{3}\sqrt{x} \text{ ms}^{-1}$ will be the minimum velocity required by a rocket to pull out of gravitational force of P, where v_e is escape velocity on earth. The value of x is

- (1) 2
- (2) 3
- (3) 18
- (4) 1

Space for your notes:

Q17 - 01 February - Shift 2

For a body projected at an angle with the horizontal from the ground, choose the correct statement.

- (1) Gravitational potential energy is maximum at the highest point.
- (2) The horizontal component of velocity is zero at highest point.
- (3) The vertical component of momentum is maximum at the highest point.
- (4) The kinetic energy (K.E.) is zero at the highest point of projectile motion.

Space for your notes:

Q18 - 01 February - Shift 2

The escape velocities of two planets A and B are in the ratio 1 : 2. If the ratio of their radii respectively is 1 : 3, then the ratio of acceleration due to gravity of planet A to the acceleration of gravity of planet B will be:

- (1) $\frac{4}{3}$
- (2) $\frac{3}{2}$
- (3) $\frac{2}{3}$
- (4) $\frac{3}{4}$

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (4)

Q2 (3)

Q3 (3)

Q4 (3)

Q5 (2)

Q6 (2)

Q7 (3)

Q8 (1)

Q9 (4)

Q10 (4)

Q11 (2)

Q12 (4)

Q13 (2)

Q14 (2)

Q15 (3)

Q16 (1)

Q17 (1)

Q18 (4)

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Q1 (4)

Acceleration due to gravity at height h

$$g' = \frac{g}{\left[1 + \frac{h}{R}\right]^2}$$

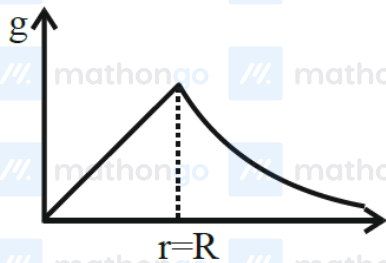
So weight at given height

$$mg' = \frac{mg}{\left[1 + \frac{h}{R}\right]^2} = \frac{18}{\left[1 + \frac{1}{2}\right]^2} = 8\text{N}$$

Q2 (3)

$$g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2}$$

$$g' = g \left\{1 - \frac{d}{R}\right\}$$



Statement I is correct & Statement II is incorrect

Q3 (3)

$$T^2 \propto R^3 \Rightarrow \left(\frac{T_1}{T_2} \right)^2 = \left(\frac{R_1}{R_2} \right)^3$$

$$\Rightarrow \left(\frac{1}{2.83} \right)^2 = \left(\frac{1.5 \times 10^6}{R_2} \right)^3$$

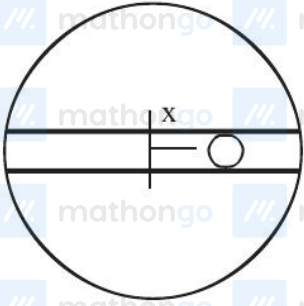
$$\Rightarrow R_2 = \left[(2.83)^2 \times (1.5 \times 10^6)^3 \right]^{1/3}$$

$$= 8^{1/3} \times 1.5 \times 10^6 = 3 \times 10^6 \text{ km}$$

Q4 (3)

$$T \propto \frac{1}{\sqrt{g}}$$

Q5 (2)



Let at some time particle is at a distance x from centre of Earth, then at that position field

$$E = \frac{GM}{R^3} x$$

$\therefore$ Acceleration of particle

$$\vec{a} = -\frac{GM}{R^3} \vec{x}$$

$$\Rightarrow \omega = \sqrt{\frac{GM}{R^3}} = \sqrt{\frac{g}{R}}$$

$$\text{Now } T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{R}{g}}$$

$$\Rightarrow T = 2 \times 3.14 \times \sqrt{\frac{6400 \times 10^3}{10}}$$

$$= 2 \times 3.14 \times 800 \text{ sec} \approx 1 \text{ hour } 24 \text{ minutes}$$

Q6 (2)

At surface of earth time period

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

At height $h = R$

$$g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2} = \frac{g}{4}$$

$$\therefore xT = 2\pi\sqrt{\frac{\ell}{(g/4)}}$$

$$\Rightarrow xT = 2 \times 2\pi\sqrt{\frac{\ell}{g}}$$

$$\Rightarrow xT = 2T \Rightarrow x = 2$$

Q7 (3)

$$U = \frac{-GM_em}{r}$$

$$U_i = \frac{-GM_em}{R_e}$$

$$U_f = \frac{-GM_em}{(R_e + h)} = \frac{-GM_em}{R_e + 2R_e}$$

$$\frac{-GM_em}{3R_e}$$

Increase in internal energy $\Delta U = U_f - U_i$

$$= \frac{2}{3} \frac{GM_em}{R_e}$$

$$\frac{2}{3} \frac{GM_e}{R_e^2} m R_e$$

$$= \frac{2}{3} mgR_e$$

Q8 (1)

$$F = \frac{Gm_1m_2}{r^2}$$

$$\Rightarrow F \propto \frac{1}{r^2}$$

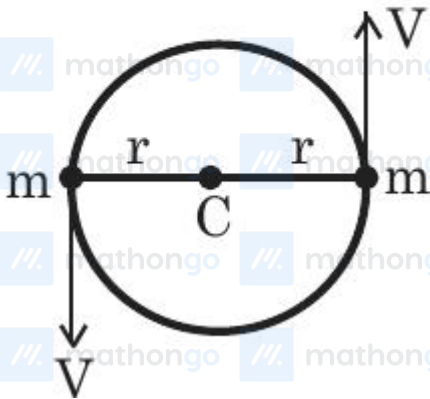
$$\Rightarrow F \propto m_1m_2$$

$\Rightarrow$ This force provides centripetal force and acts towards sun

$$\Rightarrow T^2 \propto a^3 \text{ (Kepler's third law)}$$

Q9 (4)

$$\frac{Gm^2}{4r^2} = \frac{mv^2}{r}$$



Q10 (4)

$$T^2 \propto R^3$$

$$\frac{T_1^2}{T_2^2} = \frac{R_1^3}{R_2^3} \Rightarrow \left(\frac{T_1}{T_2}\right)^2 = \left(\frac{R}{\frac{R}{4}}\right)^3$$

$$\therefore \frac{T_1^2}{T_2^2} = 64$$

$$\therefore T_2^2 = \frac{T_1^2}{64}$$

$$\therefore T_2 = \frac{24}{8} = 3$$

Q11 (2)

$$-\frac{dV}{dr} = -\frac{k}{r^2} \Rightarrow \int_{10}^V dV = \int_2^3 \frac{k}{r^2} dr$$

$$V - 10 = k \left[\frac{1}{2} - \frac{1}{3} \right]$$

$$V - 10 = \frac{k}{6} \Rightarrow V = 11 \text{ volts}$$

Q12 (4)

$$\frac{GM}{R^2} \left[1 - \frac{d}{R} \right] = \frac{4 \times GM}{(4R)^2}$$

$$1 - \frac{d}{R} = \frac{1}{4} \Rightarrow \frac{d}{R} = \frac{3}{4} \Rightarrow d = \frac{3}{4}R$$

$$d = 4800 \text{ km}$$

Q13 (2)

When metal is passing through magnetic field, eddy current will produce and it will oppose the motion, so it will take more time.

Q14 (2)

No Solution

Q15 (3)

$$g_{\text{eff}} = g - \omega^2 R_e \sin^2 \theta, \theta \rightarrow \text{co-latitude angle}$$

$$g_{\text{eff}} = g \left(1 - \frac{d}{R_e} \right), d \text{ here depth}$$

Q16 (1)

$$v_{(\text{escape})\text{planet}} = \sqrt{\frac{2GM_P}{R_P}}$$

$$= \sqrt{\frac{2G\left(\frac{M_e}{9}\right)}{\left(\frac{R_e}{2}\right)}} = \frac{v_e \sqrt{2}}{3} \therefore x=2$$

Q17 (1)

At highest point

$$V_y = 0$$

$$V_x = u_x = u \cos \theta$$

$$U_g = mgh, \text{ it is maximum at } H_{\text{max}}$$

Q18 (4)

$$V_e = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2G\rho \frac{4}{3}\pi R^3}{R}} = C\sqrt{\rho \cdot R}$$

$$\frac{V_{e_1}}{V_{e_2}} = \frac{R_1}{R_2} \sqrt{\frac{\rho_1}{\rho_2}} = \frac{1}{2}$$

$$\frac{R_1^2}{R_2^2} \times \frac{\rho_1}{\rho_2} = \frac{1}{4}$$

$$\frac{R_1}{R_2} = \frac{1}{3}$$

$$g = \frac{GM}{R^2} = \frac{G \frac{4}{3}\pi R^3 \times \rho}{R^2} = C \cdot \rho R$$

$$\frac{g_1}{g_2} = \frac{\rho_1 R_1}{\rho_2 R_2} = \frac{1}{4} \frac{R_2^2}{R_1^2} \times \frac{R_1}{R_2}$$

$$= \frac{1}{4} \times \frac{R_2}{R_1} = \frac{3}{4}$$